

CUSTOMER CHURN & COHORT RETENTION ANALYTICS

Master Technical & Behavioral Interview Guide | Candidate: Aditya Singh Bisen

Target Role: Data Analyst / BI Analyst | Stack: SQL, Power BI, DAX, Python

SECTION 1: PROJECT OVERVIEW & ELEVATOR PITCH

Q1: Can you walk me through this Customer Churn & Cohort Retention project?

In Simple Words: Give a 60-second summary: business problem, tech used, findings, and financial impact.

How to Answer in Interview: "In this project, I analyzed over 5,000 SaaS customers and 21,000 monthly transactions to address customer churn and revenue loss. Using SQL CTEs and window functions, I built a 12-month cohort retention matrix to track customer drop-offs over time. In Power BI, I engineered an executive dashboard measuring Active vs. Lost MRR, Churn Rate %, and Customer Lifetime Value (CLV). My analysis revealed an 18% churn cliff at Month 3 and found that customers logging 3+ support tickets churned at 68%. By targeting these pain points, I formulated retention strategies that projected to save ~,000 in recurring revenue."

Pro Tip: End your answer with the measurable business impact (,000 saved) to immediately catch the interviewer attention.

Q2: Why did you choose this project and what business problem does it solve?

In Simple Words: Retaining existing customers is 5x cheaper than acquiring new ones. Churn directly kills cash flow.

How to Answer in Interview: "Customer retention is the lifeblood of any subscription or e-commerce business. While acquiring customers is expensive, reducing churn directly protects monthly cash flow. I chose this project because it solves real commercial problems: identifying WHEN users drop off, WHY they leave, and WHICH acquisition channels bring the most loyal customers."

Pro Tip: This shows strong business acumen, not just theoretical coding.

SECTION 2: CORE METRICS & BUSINESS DEFINITIONS

Q3: What is Customer Churn and how did you calculate Churn Rate?

In Simple Words: Churn means customers cancelling. The rate is churned customers divided by total customers.

How to Answer in Interview: "Customer churn happens when a paying subscriber cancels their plan or stops purchasing. I calculated Churn Rate as the number of churned customers divided by total customers acquired, multiplied by 100. Across our 5,000 customer base, approximately 1,420 churned, resulting in an annualized churn rate of ~28.4%."

Pro Tip: Mention that churn can be tracked by customer count (Logo Churn) or by revenue lost (Revenue Churn).

Q4: What is Cohort Analysis and why is it better than simple monthly churn?

In Simple Words: A cohort groups users by their signup month. It tracks their survival over time.

How to Answer in Interview: "Simple monthly churn gives a single blended average, which can hide early drop-offs. In Cohort Analysis, we group customers by their initial signup month (e.g., January 2024 cohort) and track that specific group across Month 0, Month 1, Month 2, up to Month 11. This helps us pinpoint exactly at what point in the customer journey users abandon the product."

Pro Tip: Interviewers consider cohort analysis a benchmark skill for modern product and business analysts.

Q5: What is a Cohort Index?

In Simple Words: Cohort Index is simply the month counter (0 for signup month, 1 for month after, etc.).

How to Answer in Interview: "The Cohort Index is an integer representing the number of months elapsed since the customer first signed up. Month 0 is their signup month (retention is 100%). Month 1 is their second month, Month 2 is their third month, and so on. In SQL, I calculated this by finding the difference in years and months between the transaction date and their initial signup date."

Pro Tip: Formula: $(\text{Transaction_Year} - \text{Cohort_Year}) * 12 + (\text{Transaction_Month} - \text{Cohort_Month})$.

Q6: What is MRR, and what is the difference between Active MRR and Lost MRR?

In Simple Words: MRR is predictable monthly revenue. Active is money coming in; Lost is money lost to churn.

How to Answer in Interview: "MRR stands for Monthly Recurring Revenue—the predictable revenue expected every month. Active MRR represents cash coming from active subscribers. Lost MRR represents recurring revenue lost due to cancellations. Tracking Lost MRR shows leadership the exact financial impact of churn."

Pro Tip: Lost MRR is also called Revenue Churn.

Q7: What is Customer Lifetime Value (CLV) and Net Revenue Retention (NRR)?

In Simple Words: CLV is total revenue from one customer. NRR measures revenue retained from existing users.

How to Answer in Interview: “Customer Lifetime Value (CLV) is the average total revenue a company earns from a single customer over their lifespan. Net Revenue Retention (NRR) measures the percentage of recurring revenue retained from existing customers, accounting for upgrades, downgrades, and cancellations. An NRR above 100% means the business grows organically even without acquiring new customers.”

Pro Tip: Enterprise tier had the highest CLV (~,200+) due to annual billing and lower churn.

SECTION 3: SQL TECHNICAL QUESTIONS & QUERY LOGIC

Q8: Can you explain the step-by-step SQL logic used for the cohort retention matrix?

In Simple Words: Explain the 4 CTEs: find signup month -> calculate month offset -> find baseline size -> compute %.

How to Answer in Interview: “I structured the query using 4 CTEs: First (customer_cohort), I used MIN(transaction_date) to find each user initial signup month. Second (monthly_activity), I joined transactions with the cohort table and computed the Cohort Index. Third (cohort_size), I counted initial users at Month 0. Fourth (retention_counts), I counted distinct active users for each cohort month and cohort index. Finally, I divided active users by initial cohort size to produce the retention percentage.”

Pro Tip: Naming CTEs logically shows structured problem-solving.

Q9: Why did you use CTEs (WITH clause) instead of nested subqueries?

In Simple Words: CTEs make complex multi-step queries clean, readable, and easy to debug.

How to Answer in Interview: “Because cohort analysis involves multiple analytical steps, nested subqueries would be 4-5 layers deep, making the code unreadable and hard to maintain. CTEs break the query into modular, sequential blocks that execute cleanly and make peer review easier.”

Pro Tip: Shows you care about production-quality code and team maintainability.

Q10: Which SQL window functions did you use in this project?

In Simple Words: Window functions compute values across sets of rows without collapsing row details.

How to Answer in Interview: “I used MIN(transaction_date) OVER(PARTITION BY customer_id) to dynamically assign cohort signup dates. I also utilized DENSE_RANK() and LAG() to inspect payment intervals and identify customers who skipped consecutive billing cycles before churning.”

Pro Tip: DENSE_RANK and LAG are standard interview questions—highlighting them proves practical ability.

SECTION 4: POWER BI & DAX MODELING QUESTIONS

Q11: How did you model the data in Power BI? Explain your star schema.

In Simple Words: Fact_Transactions in the center, Dim_Customers and Dim_Date linked via 1-to-many relationships.

How to Answer in Interview: “I implemented an optimized Star Schema. The central Fact_Transactions table holds granular billing events like amount and date. Surrounding it are Dim_Customers (holding customer tier, channel, usage, and churn status) and Dim_Date (for time intelligence), linked by 1-to-many relationships. This prevents circular dependencies and optimizes DAX engine performance.”

Pro Tip: Always mention Star Schema as best practice over flat denormalized tables.

Q12: Can you explain the DAX formula for Cohort Retention %?

In Simple Words: Use ALLEXCEPT to lock initial cohort baseline, then divide current active by baseline.

How to Answer in Interview: “The formula uses two variables: InitialCohortUsers uses CALCULATE(DISTINCTCOUNT(Fact_Transactions[customer_id]), ALLEXCEPT(Fact_Transactions, Fact_Transactions[cohort_month])). ALLEXCEPT clears the column filter on the cohort index so it always captures the Month 0 baseline size. The second variable counts active users in that cell, and DIVIDE gives the percentage.”

Pro Tip: Demonstrating how ALLEXCEPT modifies filter context proves real DAX competence.

Q13: How did you build the Cohort Heatmap in Power BI?

In Simple Words: A Matrix visual with Cohort Month on rows, Cohort Index on columns, and conditional formatting gradient.

How to Answer in Interview: “I inserted a Matrix Visual, placing cohort_month on Rows and cohort_index (0 to 11) on Columns, with [Cohort Retention %] in Values. In Cell Elements, I enabled Background Color Conditional Formatting with a 3-point color gradient: soft red/yellow for low retention (<40%) up to dark navy blue for 100% retention.”

Pro Tip: Visual heatmaps make cohort decay instantly readable for executives.

SECTION 5: BUSINESS INSIGHTS & STRATEGIC RECOMMENDATIONS

Q14: What was the Month 3 Cliff you discovered in your data?

In Simple Words: Retention was steady for Months 1 and 2, but dropped sharply by 18% at Month 3.

How to Answer in Interview: “The cohort matrix showed retention was steady around 88-92% during the first two months. However, in Month 3, retention dropped sharply by 18%. By cross-referencing customer usage logs, I found that users who did not adopt secondary features within their first 60 days felt the product was too expensive and cancelled once their initial onboarding discount expired.”

Pro Tip: This proves you investigate root causes, not just report surface metrics.

Q15: What is the connection between support tickets and customer churn?

In Simple Words: Users with 3+ tickets churned at 68% due to unresolved technical friction.

How to Answer in Interview: “I grouped customers into 4 tiers: 0 tickets, 1-2 tickets, 3-4 tickets, and 5+ tickets. Customers with 0 to 2 tickets had a healthy churn rate of under 14%. But customers who submitted 3 or more tickets had a 68.2% churn rate. This proved that unresolved technical friction and support response delays were direct leading indicators of churn.”

Pro Tip: Actionable recommendation: Create an automated alert for Customer Success when an account logs its 2nd ticket.

Q16: Which customer acquisition channel performed the worst, and what was your recommendation?

In Simple Words: Affiliate Partners and Google Ads had high churn; LinkedIn and Organic had strong retention.

How to Answer in Interview: “Affiliate Partners had the highest churn rate at 36.2%, followed by Google Ads at 31.8%. In contrast, LinkedIn Referrals had only an 18.4% churn rate and the highest average customer lifespan. I recommended reallocating 30% of the affiliate marketing budget into LinkedIn and organic community building to acquire higher-intent, stickier customers.”

Pro Tip: Analyzing marketing channel efficiency is a high-value skill in product analytics.

Q17: How did you calculate the ~,000 revenue recovery figure?

In Simple Words: Retaining just 15% of Month-3 drop-offs preserves their recurring fee across the year.

How to Answer in Interview: “Approximately 260 customers were churning specifically at the Month 3 cliff, accounting for ~,500 in lost monthly recurring revenue. I modeled an intervention strategy: if proactive onboarding check-ins and adoption guides retained just 15% of those at-risk accounts (~39 customers across Pro and Enterprise tiers), it preserves ~,000/month, resulting in ~,000 in saved annual recurring revenue.”

Pro Tip: Being able to walk through your ROI math gives hiring managers immense confidence in your work.